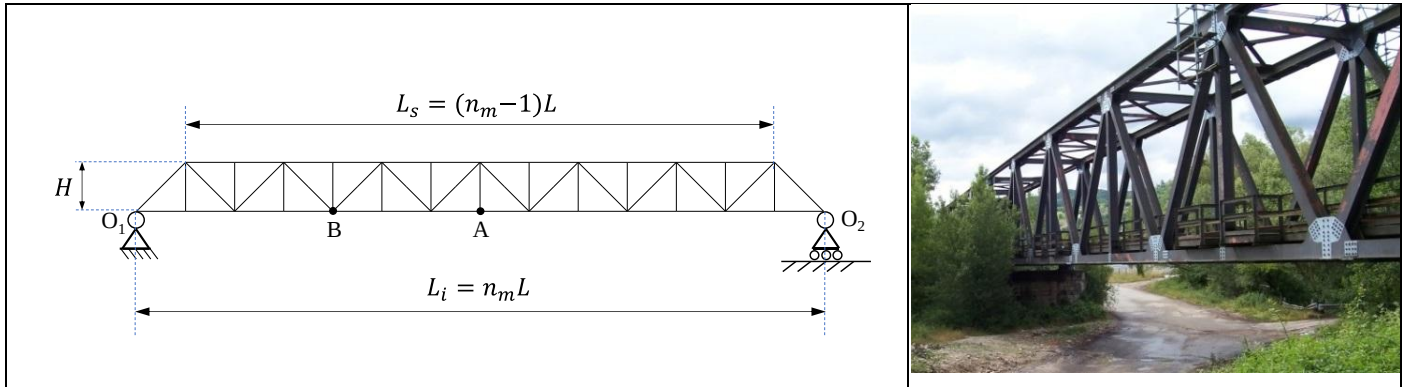


## Single span metallic truss bridge

The figure below on the left shows a plane model of a single-span metallic truss bridge, like the one shown on the right.



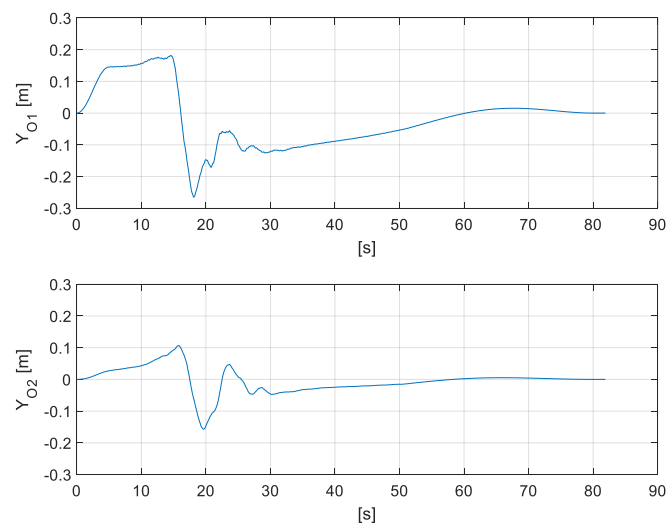
The bridge constraints are a hinge in  $O_1$  and a cart in  $O_2$ . Structural damping is to be accounted for according to the proportional damping assumption,  $[C] = \alpha[M] + \beta[K]$ , with values  $\alpha$  and  $\beta$  assigned in the input data table.

You are asked to:

- 1) Define a finite element model valid in the frequency range  $0 \div 30$  Hz (use 1.5 as the safety coefficient to define the maximum length allowed for the beam elements).
- 2) Compute the system's natural frequencies and related modes of vibration in the frequency range  $0 \div 30$  Hz.
- 3) Compute the following frequency response functions (FRF) in the frequency range  $0 \div 30$  Hz with step 0.01 Hz:
  - vertical displacement of point A produced by a vertical force on point A
  - vertical displacement of point B produced by a vertical force on point A
  - vertical acceleration of point A produced by a vertical force on point B
  - vertical acceleration of point B produced by a vertical force on point B

Point A is at mid-span, while the distance between point B and  $O_1$  is  $2L$ .

- 4) Compute the bridge response due to a seismic motion of the ground, represented as a vertical displacement of points  $O_1$  e  $O_2$  shown on the figure below:



The time histories of the vertical displacements at points  $O_1$  and  $O_2$  are given in the *seismic\_displ.txt* file available in WeBeep. The file has three columns reporting the time, the displacement at  $O_1$  and the displacement at  $O_2$  respectively.

You are requested to plot:

- The spectrum of the input displacements  $y_{O1}$ ,  $y_{O2}$ ;
- The spectrum of the vertical displacements of points A and B;
- The time histories of the vertical displacements of points A and B;
- The spectrum and the time histories of the vertical accelerations of points A and B.

- 5) Considering the passage with constant speed  $V$  of a sequence of moving concentrated loads with distance of 26 m one to the other, discuss the possibility of producing a resonance condition in the bridge for specified values of train speed  $V$ . To this end, consider an infinite sequence of moving load (approximation of a long train). Hint: use the modal superposition approach to answer this question.
- 6) Define a structural change of the bridge that allows for a 15% reduction of the maximum amplitude of the vertical acceleration of point B produced by a vertical force on point B. The maximum allowed increase of the total mass of the bridge is +5%. It is not allowed to introduce additional constraints acting on the structure.

#### INPUT DATA:

*Geometric data:*

|                                     |     |     |
|-------------------------------------|-----|-----|
| Horizontal length of one module $L$ | [m] | 10  |
| Modules number $n_m$                |     | 7   |
| Bottom chord total length $L_b$     | [m] | 70  |
| Top chord total length $L_s$        | [m] | 60  |
| Bridge height $H$                   | [m] | 4.0 |

*Inertial and elastic properties of the top and bottom chord*

|                    |                    |        |
|--------------------|--------------------|--------|
| Cross section type |                    | IPE500 |
| A                  | [cm <sup>2</sup> ] | 115.5  |
| J                  | [cm <sup>4</sup> ] | 48200  |

*Inertial and elastic properties of the diagonal beams*

|                    |                    |        |
|--------------------|--------------------|--------|
| Cross section type |                    | HEA300 |
| A                  | [cm <sup>2</sup> ] | 112.5  |
| J                  | [cm <sup>4</sup> ] | 18260  |

*Inertial and elastic properties of the vertical beams*

|                    |                    |        |
|--------------------|--------------------|--------|
| Cross section type |                    | IPE240 |
| A                  | [cm <sup>2</sup> ] | 76.84  |
| J                  | [cm <sup>4</sup> ] | 7763   |

*Material: steel - Coefficients for the structural damping evaluation*

|          |                      |         |
|----------|----------------------|---------|
| $\rho$   | [km/m <sup>3</sup> ] | 7800    |
| $E$      | [N/m <sup>2</sup> ]  | 2.06E11 |
| $\alpha$ | [s]                  | 0.15    |
| $\beta$  | [s <sup>-1</sup> ]   | 9.0e-5  |